

Extended Syllabus

(The details of the syllabus are subject to change)

Course Title	Fundamentals of Compiler Construction (기초컴파일러구성)	Semester	2 nd Semester, 2025
Credit	3	Course Number	CSE4120
Class Time	Wed/Fri 15:00~16:15	Enrollment Eligibility	3 rd ~4 th grade (recommended)

	Instructor: Jaeseung Choi (최재승)	Homepage: islab-sogang.github.io
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	Office: AS 711 Office hour: Will be announced in the first class	

I. Course Overview

1. Description

In this course, students will learn about the design and implementation of compiler. The course will mainly focus on the first two phases of a compiler: (1) front-end (for parsing) and (2) middle-end (for optimization). The back-end (code generation) part will be briefly covered at the end of the course. Students will implement their own compiler during the course project.

2. Prerequisites

Although there is no **mandatory prerequisite course**, the course will be easier if you are familiar with various fields of computer science. For instance, the front-end of a compiler is closely related to **automata theory**, and the middle-end is closely related to **programming languages**.

In this course, you will learn and use **F#** for the project. The course will provide a tutorial and exercise for **F#**, so it is okay even if you have not used it before. However, you must have sufficient programming experience.

3. Course Format

Lecture	Experiment
90%	10%

4. Evaluation

Midterm Exam	Final Exam	Assignment	Project
35%	20%	10%	35%

II. Course Objectives

By learning the fundamental principles of a compiler, students will learn how a programming language is internally implemented. Also, the students will get familiar with program analysis theories by exploring the optimization techniques of a compiler. Lastly, the projects will provide students with an opportunity to practice developing large and complex software.

III. Course Format

Every week we will have two lectures. The lecture will cover fundamental principles about compiler. This course has several challenging projects, so the lecture will sometimes give you advices and feedback on these projects.

IV. Course Requirements and Grading Criteria

There is no requirement for completing this course, as long as you don't get FA. Cut-off lines for each grade will be decided during the semester, considering the difficulty of exams and projects.

V. Course Policies

As the course mainly relies on lecture, it is strictly prohibited to disrupt the class (for example, by chatting loudly with your friends or wandering around during the class). And the course will follow the attendance policy of Sogang University.

Cheating in the exam will result in immediate **F** grade. Code copy is also strictly forbidden, and you will get zero point and your final letter grade will be downgraded (e.g., **A0** -> **B0**).

VI. Materials and References

The course will primarily rely on the lecture notes provided from the course. Interested students may refer to *"Compilers: Principles, Techniques, and Tools"* to deepen their understanding, but it is not mandatory.

VII. Course Schedule (subject to change)

For all of the weeks, (1) the class method will take lecture format, and (2) the lecture notes uploaded in *Cyber Campus* will be used as the class material.

Week 1	Learning Objectives	Learn the overall structure of a compiler
	Topics	Orientation, course topic overview
	Assignments	
Week 2	Learning Objectives	Learn the background for lexical analysis
	Topics	Regular expression, Non-deterministic FA
	Assignments	
Week 3	Learning Objectives	Learn the key principles of lexical analysis
	Topics	Deterministic FA
	Assignments	
Week 4	Learning Objectives	Learn the background for lexical analysis
	Topics	Context-free grammar
	Assignments	
Week 5	Learning Objectives	Learn the key principles of syntax analysis
	Topics	Top-down parsing
	Assignments	
Week 6	Learning Objectives	Learn the key principles of syntax analysis
	Topics	Bottom-up parsing, parser generator (recorded video lecture)
	Assignments	Assignment #1: Lex / Yacc (Flex / Bison)
Week 7	Learning Objectives	Learn the advanced topics of syntax analysis
	Topics	Bottom-up parsing (cont.)
	Assignments	

Week 8	Midterm exam period	
Week 9	Learning Objectives	Prepare the course project
	Topics	Exam review, F# tutorial, Project overview
	Assignments	Assignment #2: F# exercise
Week 10	Learning Objectives	Learn the key principles of type checking for C-like languages
	Topics	Type checking
	Assignments	Project phase #1: Type checking
Week 11	Learning Objectives	Learn the key principles of IR generation
	Topics	IR generation
	Assignments	
Week 12	Learning Objectives	Learn the background for IR-level optimization
	Topics	Basics of optimization
	Assignments	Project phase #2: IR generation
Week 13	Learning Objectives	Learn the key principles of IR-level optimization
	Topics	Various optimizations (CP, DCE, Mem2Reg, etc.)
	Assignments	
Week 14	Learning Objectives	Learn the advanced topics of IR-level optimization
	Topics	Advanced optimizations, register allocation
	Assignments	Project phase #3: Optimization
Week 15	Learning Objectives	Review the topics covered in the course
	Topics	Final exam, course wrap-up
	Assignments	
Week 16	Final exam period	

VIII. Special Notes

Course announcements will be made in Cyber Campus (cyber.sogang.ac.kr).

IX. Aid for the Challenged Students

The instructor will sincerely respond to the request from *Support Center for Students with Disabilities* (장애학생지원센터). Students with disabilities may also contact the instructor and ask for helps, such as an allocation of a front seat.